Macromolecules

pubs.acs.org/Macromolecules Article

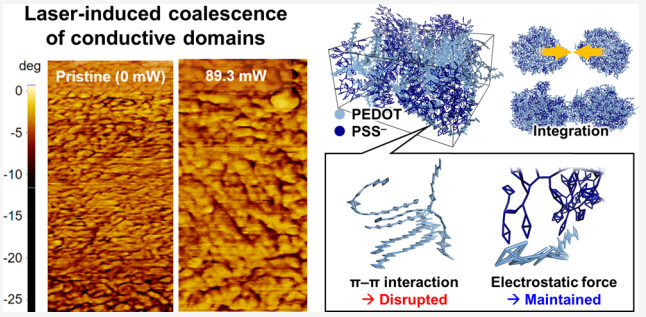
Thermodynamic Mechanism Governing the Coalescence of Conductive Particles in PEDOT:PSS under Laser Irradiation

Hongdeok Kim, Daeyeon Won, Seung Hwan Ko,* and Joonmyung Choi*

Cite This: *Macromolecules* 2024, 57, 2048−2056 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: The strategy of obtaining selective electrical properties by laser irradiation of poly(3,4-ethylenedioxythiophene) stabilized with poly(4-styrenesulfonate) (PEDOT:PSS) is breaking new ground in the fabrication process of conductive films. In this study, the theoretical mechanism by which the laser irradiation induces coalescence of PEDOT-rich particles and improves electrical properties of PEDOT:PSS was explored from a thermodynamic perspective. The microstructural evolution by laser irradiation was experimentally verified, and the equivalent environment was implemented from coarse-grained molecular dynamics simulations. The results suggest that selective disruption between $\pi-\pi$ interactions and electrostatic attraction, the forces governing the PEDOT-rich domain, is a key driving factor for enhanced phase separation in PEDOT:PSS. The energy supplied from the laser provides sufficient molecular mobility to allow the PEDOT-rich core to integrate with other adjacent cores without compromising the strong internal cohesion of its own core. The enhanced mobility leads to bridging-generating events with triggering of active molecular rearrangements between the cores, driving the coalescence of conductive particles.



1. INTRODUCTION

Conductive polymer has recently received considerable attention due to its potential applications in low-cost, flexible, and printable soft electronic devices.[1−6] One of the most promising is poly(3,4-ethylenedioxythiophene) stabilized with poly(4-styrenesulfonate), also known as PEDOT:PSS. PEDOT:PSS is emerging as the core of conductive polymers due to its excellent electrical conductivity, chemical stability, elasticity, and facile solution processing.[7−9] In particular, it exhibits remarkable structural stability even in aqueous or physiological environments, making it a promising candidate for bioelectronics applications such as implants for real-time monitoring of biological activity or devices for human−machine interfaces.[10−13]

The outstanding functionality of PEDOT:PSS is attributed to its unique morphology. From a microscopic point of view, PEDOT:PSS has a core−shell structure in which spherical PEDOT-rich particles with a diameter of 10−50 nm is surrounded by PSS-rich shell.[14−17] Because the conductive PEDOT-rich phase is completely encapsulated in nonconductive domains, pristine PEDOT:PSS films exhibit very low electrical conductivity (∼1 S/cm).[18−20] In order to secure substantial conductivity, therefore, additional physicochemical treatment is required to induce connections between PEDOT-rich domains. Typical chemical treatment methods include the addition of polar solvents such as ethylene glycol (EG), sulfuric acid (H_2SO_4), and dimethyl sulfoxide (DMSO).[21−24] Polar solvents disrupt the ionic bonding between PSS and PEDOT while dissolving the PSS molecules during immersion, thus facilitating the penetration of PEDOT into the PSS-rich shell.[25,26] However, the high cytotoxicity of most polar solvents significantly restricts their application, particularly in bioelectronics.

Recently, laser-induced microannealing has emerged as a nonchemical method to improve the electrical properties of PEDOT:PSS.[10,27−29] Laser irradiation dramatically elevates electrical conductivity up to 932 S/cm, surpassing the capability achievable with conventional chemical treatments.[28] A key mechanism arises from the high absorption of PEDOT for laser photons. The selectively energized PEDOT molecules merge adjacent PEDOT-rich phases, facilitating further separation of the individual phase at the microscale and forming efficient charge transport pathways throughout the film.[30] Such laser processing has unique advantages in that it does not require additional chemicals and allows more precise control of the level of phase splitting by varying the laser intensity and speed. In addition, spatially selective laser irradiation enables the design of high-resolution conductive

Received: September 13, 2023
Revised: December 19, 2023
Accepted: February 8, 2024
Published: February 21, 2024



ACS Publications © 2024 American Chemical Society

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

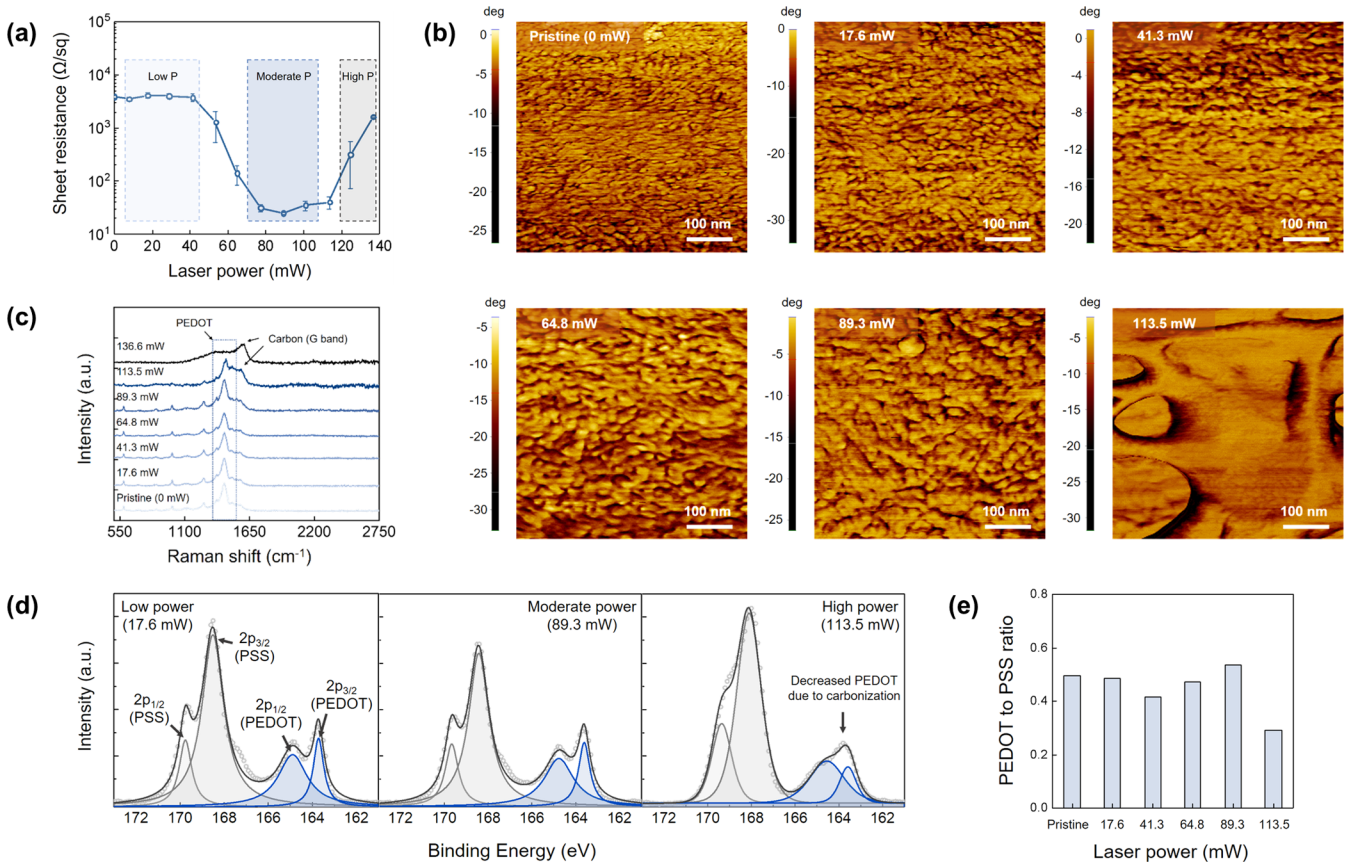


Figure 1. Laser-induced coalescence of conductive cores in PEDOT:PSS films. (a) Changes in sheet resistance according to laser power. Each data plot represents the mean value and standard deviation for six samples. (b) Atomic force microscopy (AFM) phase images of the PEDOT:PSS film with increasing laser power. (c) Raman spectrum and (d) X-ray photoelectron spectroscopy (XPS) analysis of the treated PEDOT:PSS. (e) Relative proportion changes of PEDOT to PSS estimated from the deconvoluted spectrum of XPS.

channel patterns (~6 μm) that cannot be reached by conventional methods.[10]

Although the strategy of designing PEDOT:PSS microphases using laser has made significant engineering progress, attempts to theoretically approach the physicochemical effects of the process are still very weak. One of the major barriers is that it is very challenging to experimentally capture the activation process taking place in molecular-scale space during the short turnaround time of the laser. Indeed, the molecular-scale mechanism by which laser irradiation imparts electrical conductivity to PEDOT:PSS is still unclear. Some literature postulates cleavage of the ionic bond between PSS and PEDOT within the PEDOT-rich domain as the main driver of the enhanced separation of phases.[31] Alternative perspectives include partial ablation of nonconductive PSS domains and mutual access of PEDOT-rich cores during the laser application.[27] However, the demonstration of the dynamic change of the phase within the molecular scale remains an area of speculation until now. What is clear thus far is only the optical observation that the well-processed PEDOT-rich particles penetrate the nonconductive shell to form a three-dimensional conducting network while maintaining distinct, independent phases.[10,30]

Understanding the molecular-scale mechanism by which laser irradiation induces the reorganization of PEDOT:PSS is important for optimizing and diversifying the technology. In this regard, this study utilizes coarse-grained molecular dynamics (MD) simulations to explore the microstructure evolution of PEDOT:PSS by laser irradiation from a thermodynamic point of view. We focused on the changes in $\pi-\pi$ interactions and electrostatic force constituting the PEDOT-rich cores and demonstrated their pivotal roles as driving factors for conductive domain coalescence. In particular, the process by which a PEDOT-rich core gains mobility sufficient to integrate with other adjacent cores while maintaining strong internal cohesion was examined at the molecular scale. Therefore, this study is meaningful as a theoretical basis for the design of laser-induced polymer electrodes, which have great potential in bioelectronics, energy devices, and electronic skin applications.

2. EXPERIMENTAL DEMONSTRATION OF PHASE SEPARATION ENHANCEMENT OF PEDOT:PSS

In order to precisely capture the separating behavior of PSS and PEDOT in the microdomain, laser irradiation experiments with a wide power range were performed. The effect of laser microannealing on the PEDOT:PSS was evaluated from the electrical and morphological analysis. Sample preparation, processing, and methods of analysis are described in the Experimental Section of the Supporting Data. Figure 1a shows the sheet resistance of the PEDOT:PSS as a function of laser intensity. The profile consists of three distinct stages depending on the laser power applied. When the laser power was as low as ~41.3 mW, the resistance remained at a high level. However, as the laser power exceeds the threshold, the sheet resistance rapidly decreased, indicating a substantial improvement in electrical properties of the conducting film. The profile then

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

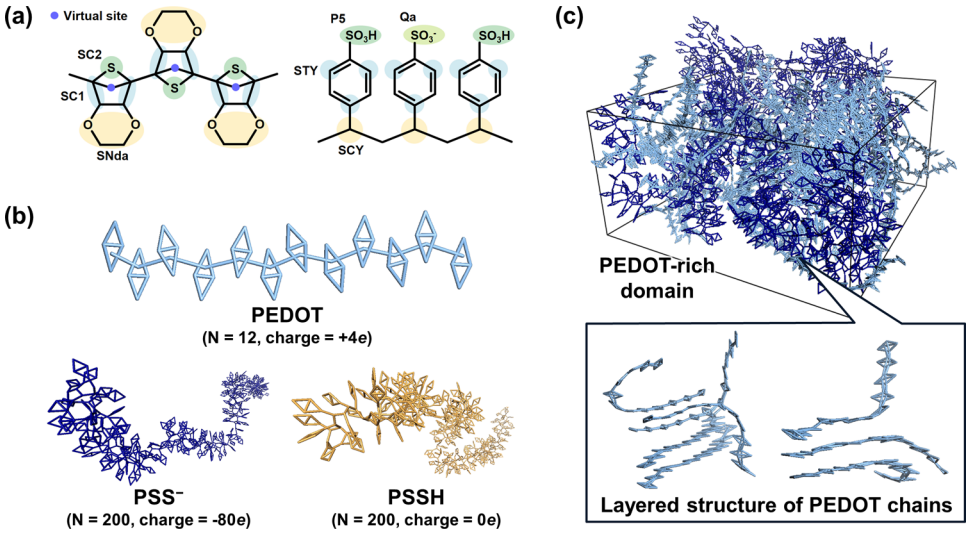


Figure 2. Coarse-grained molecular dynamics simulation modeling for laser-induced microphase changes in PEDOT:PSS. (a) MARTINI coarse-grained models for PEDOT (left) and PSS chains (right). (b) Molecular configurations of PEDOT:PSS components. PSS containing deprotonated sulfonate groups (PSS^-) and PSS containing only protonated groups (PSSH) were constructed individually. (c) Equilibrated microstructure of PEDOT-rich phase. Key snapshots of the layered structure formed by the PEDOT chains are shown in the subfigure.

forms a plateau in the power range of 64.3−100.8 mW but increases back to its initial value as the film is over-irradiated.

The aforementioned changes in electric performance by laser irradiation are closely related to the morphology of PEDOT:PSS. Figure 1b shows the atomic force microscopy (AFM) phase image under different laser intensities (topography images are also provided in Figure S1). PEDOT:PSS consist of conductive PEDOT-rich particles and a nonconductive PSS barriers separating between the particles. Due to the difference in elasticity, the PEDOT-rich (hard) and PSS (soft) microphases appear as bright and dark areas in the phase image, respectively.[32,33] In the pristine PEDOT:PSS film, the nanosized PEDOT-rich particles are extensively surrounded by PSS domains and are homogeneously distributed over the entire area due to the weak separation of microphases. This characteristic persists until the laser power reaches 41.3 mW, which coincides with the condition where the little change in electrical resistance of the sheet occurs. However, under a laser with higher intensity (64.3 and 100.8 mW), adjacent PEDOT-rich domains merge with each other and grow into larger-sized particles. As the abundant interactions between the conducting PEDOT chains are promoted, the electrical resistance of the entire microstructure is reduced to 1000th level.

At the laser power of 113.5 mW, the AFM phase image is not clear due to high surface roughness. To assess the effect of excessive laser power, we additionally obtained Raman spectra of PEDOT:PSS as shown in Figure 1c. The strongest band in the pristine PEDOT:PSS film was found at 1420 cm^{-1}, corresponding to the symmetric $C_\alpha = C_\beta$ stretching modes of the aromatic rings in the PEDOT chain.[34,35] The profile largely maintains its original shape up to the power range of 89.3 mW. However, it shifts significantly to 1590 cm^{-1}, corresponding to the G band of hexagonal carbon at intensities above 113.5 mW.[36] In other words, laser light primarily absorbed by PEDOT chains (see Figure S2) leads to bond breakage and carbonization of conductive chains. Nevertheless, the two-dimensional (2D) peak at 2680 cm^{-1} was not observed, indicating the incomplete graphenization of the polymer chains. This adequately explains the decrease in the electric conductivity of the film at excessive laser power and is consistent with the optical image results (Figures S3 and S4).

X-ray photoelectron spectroscopy (XPS) was used to measure the relative proportion changes of PEDOT to PSS during the laser process. Figure 1d shows the S(2p) spectra of the PEDOT:PSS film obtained from under different laser conditions. The profiles exhibited four independent peaks corresponding to the sulfur of the sulfonate group of PSS and the sulfur of the thiophene ring of PEDOT.[37,38] The proportion of PEDOT to PSS was quantified from the area of each peak as shown in Figure 1e. The molecular composition is maintained constant until the laser power increased to 89.3 mW. However, when the laser power is higher than 89.3 mW, the PEDOT molecule starts to decompose and the ratio is greatly reduced.

The implications of the experimental observations introduced above are straightforward. Depending on the laser power conditions, PEDOT:PSS causes aggregation of conductive particles and consequent enhanced phase separation, which is demonstrated by rapid changes in electrical performance. In particular, as the laser power gradually increased, the electrical properties rapidly improved up to a threshold point and then returned to the initial level, completely reproducing previous findings.[10,27,30] PEDOT:PSS exhibits a microstructure in which conductive PEDOT-rich cores are capsulated by nonconductive PSS-rich walls. Therefore, the electrical conductivity can change significantly depending on the cohesion and connectivity between PEDOT-rich cores, and its influence has also been claimed in reported studies.[10,27−30]

Therefore, an important question raised here is how the energy supply by such laser irradiation can induce the interconnection of the PEDOT-rich particles by what physicochemical mechanism. Two aspects are particularly noteworthy: (1) The PEDOT-rich cores maintained self-aggregation even under the injection of significant energy. (2) Despite (1), the conductive particles obtained high mobility and succeeded in forming larger clusters through mutual gathering. Due to the extremely fast processing time and microscopic physical scale of lasers, it is difficult to fully understand their mechanical principles based on the apparent properties obtained from experimental measurements. To address this, we in-depth analyze the behavior of PEDOT-rich particles surrounded by PSS domains and their energetic characteristics through coarse-grained MD simulations in the next section.

3. CHARACTERIZATION OF COALESCENCE MECHANISM OF CONDUCTIVE PARTICLES *IN SILICO*

3.1. Molecular Dynamics Simulation Setup. The coarse-grained molecular dynamics simulations were carried out using a commercial MD software, Materials Studio 2020 (Dassault Systèmes, France). The laser irradiation simulations and structural characterizations were carried out using an open-source code, LAMMPS (Sandia Lab). All the interaction

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

forces of each bead constituting an atom or molecule were described from the MARTINI force field.[39] The MARTINI force field categorized bundles of atoms into four types: polar (*P*), nonpolar (*N*), apolar (*C*), and charged (*Q*). These bead types are further subdivided based on their hydrogen-bonding capabilities (*d* for donor, *a* for acceptor, *da* for both, and 0 for none) and the degree of polarity (assigned as an integer from 1 to 5 depending on magnitude).

3.2. Molecular Modeling of PEDOT:PSS Microstructures. The molecular design of the PEDOT:PSS microstructure was inspired by the theoretical work done by Modarresi et al.[40] By simulating the drying process of solvated PEDOT:PSS, a two-phase granular structure with PEDOT- and PSS-rich regions can be identified. In particular, it is confirmed that the PEDOT-rich phase was composed of doped PEDOT chains and PSS chains with deprotonated sulfonate groups in a weight fraction of 0.46:0.54. On the other hand, PSS molecules with protonated sulfonate groups completely occupied the nonconductive domains surrounding the PEDOT-rich phase.

The chemical structures of PEDOT and PSS and the modeling process using coarse-grained beads are described in Figure 2a. The coarse-graining procedure for PEDOT chains followed the method of Modarresi et al.[41] Specifically, the backbone of PEDOT chains was represented by the linkage of virtual sites located at the center of mass of the sulfur and surrounding carbons. The virtual site has no mass and thus does not participate in van der Waals interactions. Instead, the site has a positive charge of +0.3333*e*, which equalizes the charge distribution of the PEDOT chain. Each PEDOT chain consisted of 12 repeated units, thus having a net charge of +4*e*. Meanwhile, the coarse-grained modeling of the PSS was based on the work of Vögele et al.[42] The deprotonated sulfonate group (SO_3^-) in the PSS chain was described as the bead Q_a with a negative charge of −1*e*, while the protonated one (SO_3H) was represented as the bead *P5*. The PSS chain was modeled with 200 repeat units. The net charge was set to −80*e* chain (PSS^-) and neutral for the protonated chain (PSSH), respectively. The structure of the molecular model generated from the above process is shown in Figure 2b.

Subsequently, the microstructure of the PEDOT-rich bulk phase was modeled. A total of 80 PEDOT and 4 PSS^- chains were randomly dispersed in a periodic simulation box. The weight fraction of PEDOT:PSS^- was 0.46:0.54, which was in good agreement with the theoretically known values of the PEDOT-rich phase.[40] In order to avoid undesired intermolecular interferences, the components were initially placed in a sufficiently large simulation box with the mass density of 0.1 g/cm^3. The potential energy of the entire system was minimized using a conjugate gradient algorithm with an energy cutoff of 10^{-4} kcal/mol. Subsequently, the MD simulations were performed for 10 ns under the NPT ensemble at 300 K and 1 atm conditions. The time step for the numerical integration of the equations of motion was set to 0.5 fs. After the equilibration steps were completed, the PEDOT-rich model exhibited a cell size of 88.1 × 88.1 × 44.0 Å^3 as shown in Figure 2c. Due to strong $\pi-\pi$ interactions during equilibration, PEDOT molecules spontaneously generated a layered structure composed of 3−5 chains. Such formation of layered structures by interchain interaction was consistent with the experimental and theoretical observations.[32,43]

3.3. Laser Irradiation Simulation. From a microscopic point of view, laser irradiation can be regarded as the transfer of energy from photons to the atoms of a material, resulting in a rapid and high internal energy rise in the polymer system. In this respect, we described the situation of laser irradiation by injecting additional energy into the unit cell prepared in MD simulations. The total energy induced by the laser was quantified under the following three assumptions: (1) The region irradiated by the laser has a spatially uniform energy distribution (i.e., flat top beam profile). (2) The PEDOT:PSS microstructure is considered as an isolated system in which the applied energy is not dissipated to the outside. (3) Because the characteristic length of the laser beam is sufficiently longer (4.91 μm) than the thickness of the PEDOT:PSS film (0.34 nm), the effect of the laser is uniform throughout the entire depth of the specimens. The detailed parameters for the calculation of laser-induced energy increments are given in Table S1. Referring to the experimental condition of 89.3 mW where the minimum sheet resistance of PEDOT:PSS is achieved, an equivalent volumetric energy density was supplied in the simulation box. To prevent the system from experiencing an excessively rapid increase in energy over a short period, the laser-induced energy was applied in 10 steps for a total of 1 ns. The PEDOT-rich model with additional energy applied was further equilibrated for 9 ns in the NVE ensemble.

In general, conjugated polymers (such as PEDOT in PEDOT:PSS) exhibit high phonon absorption capability over a wide wavelength range (300−1000 nm).[44] Photon absorption by the polymer causes rapid promotion of an electron from the ground state to the excited state. The excited electron subsequently returns to the ground state either by emitting a photon or by releasing thermal energy through nonradiative decay.[44] Laser conditions including wavelength and type impact the efficiency of converting electrons from the ground state to the excited state and the scale of the energy increment. However, the present study focused more on exploring the microstructural evolution under conditions where sufficient thermal energy was already generated to exceed the threshold for phase separation in PEDOT:PSS. Therefore, the influence of the differences in the efficiency and scale of the energy conversion from light to heat depending on the bonding state of organic molecules was not addressed. Notably, morphological changes and electrical property improvements have been observed for PEDOT:PSS under diverse laser conditions,[10,27−30] suggesting that the coalescence of PEDOT-rich cores can be satisfactorily explained by thermal-energy-induced structural rearrangement.

Note that the effect of the electric field induced by the laser light is excluded in the present simulation, even though it is another important factor causing the further separation of the PEDOT:PSS phases.[45,46] This is because power and duration of the electric field induced by the laser light (up to 4.63 kV/m and 20 μs) were much lower than those practically required to induce phase redistribution of PEDOT:PSS (above 300 kV/m and 10 min).[45,46] Therefore, our research focuses on the moment when the internal energy rise of the microstructure changes the morphology of PEDOT:PSS and aims to discover related morphological factors.

3.4. Microstructural Change of PEDOT-Rich Domain by Laser Irradiation. The influence of laser-induced energy on the microstructural morphology of PEDOT-rich phases was characterized. For the PEDOT-rich models equilibrated in pristine and laser irradiation states, an additional MD runs were performed for 2 ns and mean square displacement

Downloaded from pubs.acs.org/macmobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

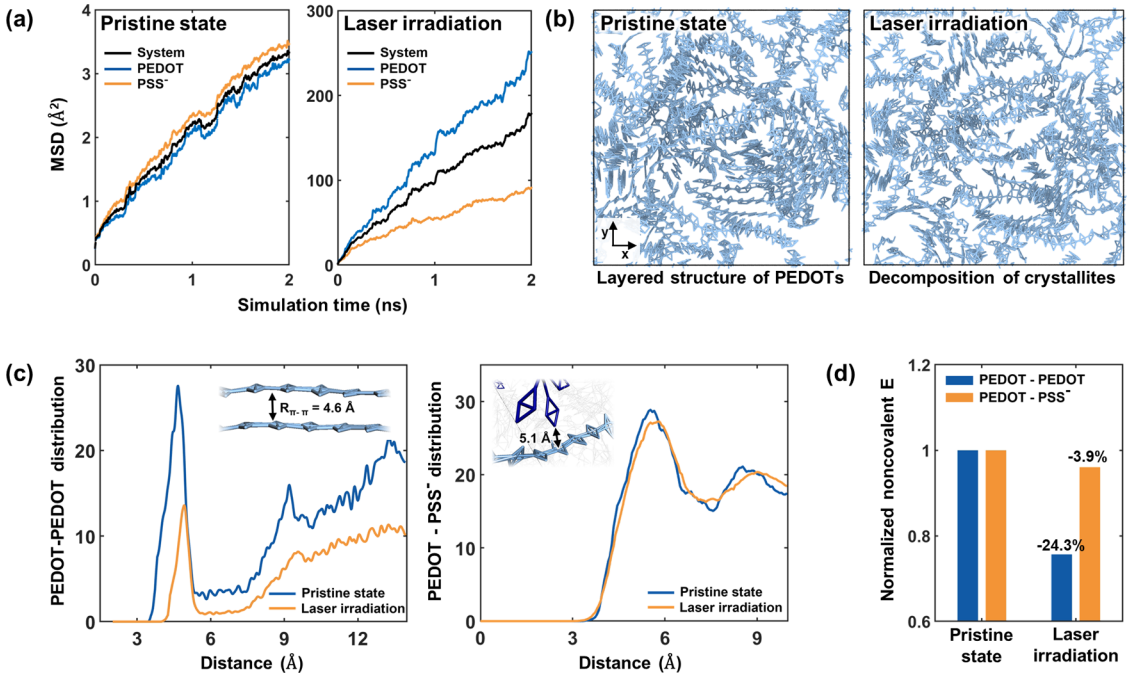


Figure 3. Morphology changes of the PEDOT-rich domains after laser irradiation in the nanoscale. (a) Mean square displacement profiles of molecular components constituting the PEDOT-rich phase. (b) Snapshots of PEDOT molecules equilibrated in pristine and laser-irradiated state. (c) Radial distribution function between virtual sites in PEDOT (left) and between *virtual sites* in PEDOT and *Qa* beads in PSS^- (right). (d) Changes in noncovalent interaction energy between the components constituting the PEDOT-rich phase. The results are presented as values relative to the pristine state energy.

(MSD) was calculated. In particular, the MSD profiles of PEDOT and PSS^- were calculated separately to determine the contribution of the components constituting the PEDOT-rich phase. As shown in Figure 3a, the laser-irradiated system shows a much steeper rise in the MSD profile compared to the pristine state. This is straightforward because the increase in kinetic energy by heating broke the rigidity of the PEDOT-rich microstructure, further increasing its mobility. What is interesting is that PEDOT and PSS^- pairs, which had similar mobilities in the pristine state, show significant differences after laser irradiation. In particular, the increase in molecular mobility is more pronounced for PEDOT than for PSS^-.

The main reason for the enhanced mobility of PEDOT chains can be found from the configuration analysis presented in Figure 3b. The high backbone rigidity and planar structure of PEDOT chains facilitate abundant $\pi-\pi$ interactions with their neighbors, generating the layered conformations throughout the microstructure. Such structural features are also clearly identified from the radial distribution function (RDF), which exhibits distinct peaks at multiples of the $\pi-\pi$ stacking distance (4.6 Å in MARTINI force field[41,47]) between the PEDOTs, as shown in Figure 3c. However, after laser irradiation, the peak values were smoothed and the overall RDF profile was significantly lowered. In other words, the alignment between the PEDOTs in the PEDOT-rich phase was greatly disrupted by the laser-induced energy. In contrast, no significant damage is observed in the interaction between PSS^- and PEDOT molecules. The RDF profile remains largely unchanged even the application of laser energy, indicating the preservation of the electrostatic attraction between PEDOT and PSS^-.

The above observations simultaneously reveal the existence and relative importance of two fundamental forces governing the PEDOT-rich phase: the electrostatic interaction force between PEDOT and PSS^- and the $\pi-\pi$ interaction force between PEDOT chains. Under the equilibrium environment, both interaction forces contribute to maintaining the structural stability of the PEDOT-rich phase. When experiencing a rapid energy increase by the laser, however, the $\pi-\pi$ interaction force between PEDOT chains is rapidly disrupted, and thus, only the electrostatic interaction between PEDOT and PSS^- dominates the entire system. The selective disruption of such interaction forces results in preferentially breaking only the layered structure of PEDOT in the PEDOT-rich phase. The results indicate that the interaction between PEDOT and PSS^- is more favorable from an energetic point of view than the interaction between PEDOT chains (see Supporting Note 1 and Figure S5 for results comparing the interaction energies of PEDOT−PEDOT and PEDOT−PSS^-). This can also be confirmed by comparing changes in noncovalent bonding forces between molecules before and after laser irradiation. As shown in Figure 3d, the rate of noncovalent interaction energy reduction between PEDOT chains (−24.3%) by laser irradiation is much more significant than that between PEDOT and PSS^- (−3.9%).

3.5. Revealing the Molecular-Level Coalescence Mechanism of PEDOT-Rich Cores. In order to further understand the aggregation process of isolated PEDOT-rich particles by laser-induced thermal energy, we prepared a coarse-grained MD model of a PEDOT:PSS system in which spherical PEDOT-rich particles are surrounded by PSS domains. First, two spherical PEDOT-rich domains are placed in the periodic unit cell with a 4.5 nm intervals along the *x*-axis. The thickness of the cell box was set to 44.0 Å, the same as the PEDOT-rich model prepared in the previous step. The simulation box shows a density of 0.1 g/cm^3 when containing only two PEDOT-rich particles. Subsequently, 40 protonated PSS (PSSH) were randomly placed in the empty space of the unit cell. Time integration was performed for 10 ns under the

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

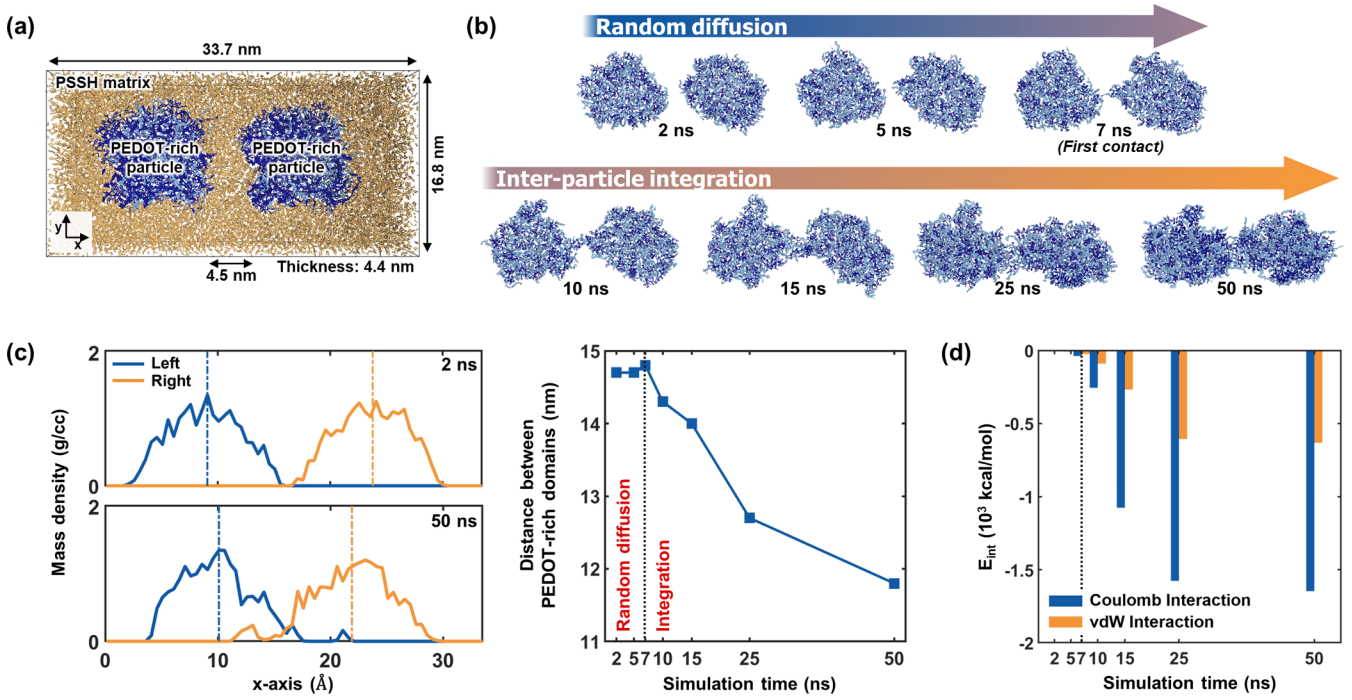


Figure 4. Molecular-scale analysis of coalescence mechanism of conductive cores under laser irradiation. (a) Coarse-grained microstructure in which two spherical PEDOT-rich particles are surrounded by a nonconductive PSSH matrix. (b) Representative snapshots of PEDOT-rich domains over simulation time. The PSSH matrix was omitted for clarity. (c) Mass distribution of PEDOT-rich constituents along the *x*-axis (left) and relative distance between the center of mass of the PEDOT-rich particles (right). In the profiles of mass distribution, the center of mass of each particle is indicated by a vertical dotted line. (d) Interaction energy between two PEDOT-rich particles over simulation time.

NVT ensemble at 300 K only for the inserted PSSH molecules, which made the PSSH molecules sufficiently surround the PEDOT-rich particles to form a stable interface. Next, time integration for 5 ns under the NVT ensemble at 300 K and time integration for 20 ns under the NPT ensemble at 300 K and 1 atm were sequentially performed for all molecules. After the model was equilibrated, the PEDOT-rich particles were completely embedded in nonconductive PSSH while remaining separated from each other, as shown in Figure 4a.

The same level of energy density as the laser irradiation condition was assigned to the PEDOT:PSS microstructure, and an additional MD run was performed for 50 ns under the NVE ensemble. The PEDOT-rich particles exhibit significant structural changes along the orientation facing each other, as shown in Figure 4b. During the initial stage of laser heating of 2−5 ns, the PEDOT-rich particles remain isolated. However, as the simulation time elapses by 7 ns, the PEDOT chains on the surface of the two separated particles come into contact with each other for the first time (see Figure S6 for key snapshots at the moment the PEDOT chains connect to each other). Such behavior is reasonable from a molecular structural point of view since PEDOT chains are glassy with high slenderness ratio and high backbone stiffness, which is advantageous for penetrating the surrounding PSSH domains. In addition, in an environment where high energy is supplied, perturbation occurs in the layered structure of the PEDOT chain, which was maintained by $\pi-\pi$ interaction, and high molecular mobility can be secured. The collapse of the layered structure resolves the intermolecular friction between the conducting chains, effectively reducing the reptation time of PEDOTs. During the subsequent time integration of ∼50 ns, the bridge width between the PEDOT-rich particles gradually increases, and the interparticle coupling becomes stronger with time. Still, no appreciable mixing is observed between the PEDOT-rich phase and the surrounding PSSH phase throughout the simulations. In other words, the PEDOT-rich particles significantly preserved their own cohesion even in high thermal energy environments. This is also in line with previous results of Figure 3c, indicating that the electrostatic forces between PEDOT and PSS^- molecules remained intact.

The approach behavior of separated particles was further characterized as shown in Figure 4c. To quantify the amount of mass exchange and its orientation during thermodynamics, the mass density of the PEDOT-rich particles was calculated at 5 Å intervals along the *x*-axis. The interparticle distance was defined relative to the center of mass of the two particles. The distance profile is almost constant at the beginning of the time integration but starts to decrease linearly after the first contact at 7 ns. This suggests that the behavior of PEDOT-rich particles approaching each other occurs in two steps. First, when the particles are spatially separated from each other and there is no interaction between them, the chains placed on the surface of the particles spread nondirectionally according to the energy supply. Second, once the formation of electrostatic interactions with neighboring particles begins, the particles are selectively attracted only in the direction they are facing each other, leading to significant coalescence of conductive cores. Such a two-step coalescence process is further supported by the noncovalent interaction energy between conductive particles presented in Figure 4d. The interaction force between the two particles is almost zero in the initial structure but forms and converges rapidly after 7 ns when the bridge by the PEDOT chain is built. In particular, due to the polarization formed by the conductive particles, the contribution of the noncovalent interaction energy is dominated by the Coulomb interaction.

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

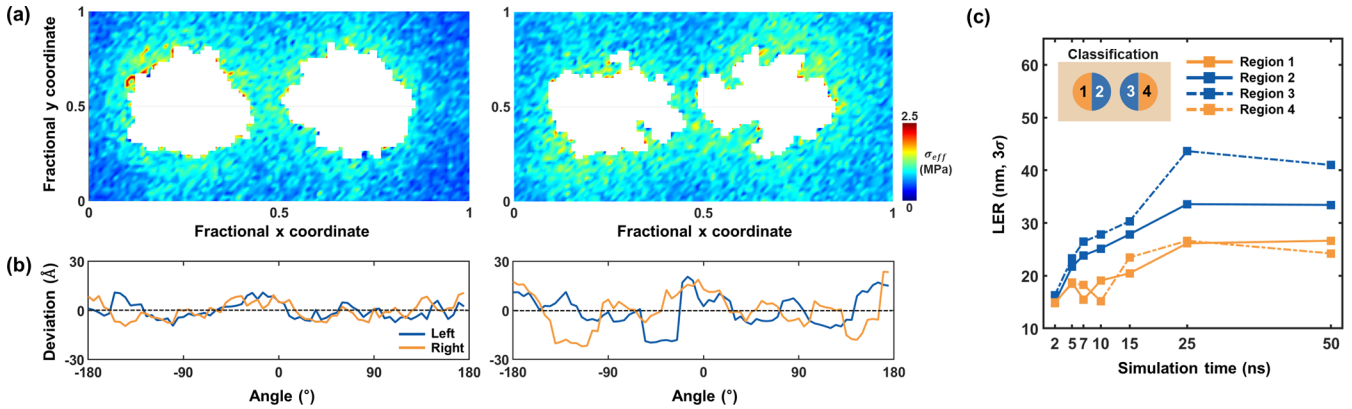


Figure 5. Disorganization of the PSSH matrix by molecular perturbation of the PEDOT-rich phase. (a) Virial stresses distributed in the PSSH matrix under laser-induced heating conditions. (b) Perimeter roughness profile of the PEDOT-rich particle. Left and right sides in panels (a, b) correspond to the initial thermodynamic state (2 ns) and the last frame (50 ns), respectively. (c) Variation of line edge roughness of the PEDOT-rich particles over simulation time.

In the previous section, it was confirmed that the electrostatic attraction between PEDOT and PSS^- is largely preserved even under laser-induced thermal energy. The finding adequately describes the property of maintaining phase separation after bridging of the PEDOT-rich particles. However, the question of how particles can achieve spontaneous mutual access in the first energized environment has not been fully resolved. In this regard, we further characterized the microstructural influence of the PSSH matrix during the thermodynamics of PEDOT-rich particles.

In order to determine the mechanical forces actually applied at the dissimilar material interface, the distribution of the effective atomic stress (σ_{eff}) generated in the PSSH matrix during laser heating was calculated as shown in Figure 5a. The stress tensor of each atom was calculated based on the virial theorem,[48] which in turn was represented as a scalar value using the von Mises criterion. This quantifies the magnitude of microscopic stress deviations formed in the surrounding PSSH matrix and thus explicitly indicates the scale of structural rearrangements (see Supporting Note 2 and Figure S7). The analysis results show that the formation of atomic stress in the PSSH domain is most prominent at the PEDOT:PSS interface. The layered structure of conductive PEDOT chains disintegrates with energy supply, and the resulting increased molecular mobility directly exchanges stress with the surrounding matrix (also see Supporting Note 3 and Figure S8 for vibration analysis). In particular, the region between the two PEDOT-rich particles is filled with atoms of higher stress magnitude than the other regions, so structural rearrangements of the molecules are easily activated. Such space-dependent differences in stress distribution still remain even in frames of 50 ns in which PEDOT-rich particles are partially integrated.

The spatial dependence of molecular rearrangements was also investigated from changes in the perimeter roughness of the PEDOT-rich particles. For the quantification of roughness values, we considered a virtual probe that scans the inner wall of the PSSH matrix based on the position of the center of mass of the PEDOT-rich particle. The roughness distribution was obtained from the change in length of the probe calculated while rotating it counterclockwise around the x-axis by 5° (see Figure S9). The deviation profiles of the perimeter roughness are presented in Figure 5b. The variation in roughness becomes more pronounced as the coalescence proceeds, suggesting that PEDOT-rich particles penetrate into the PSSH inner wall during the process. Meanwhile, Figure 5c shows the change in line edge roughness (LER) with the spatial domains of the particle over simulation time. The junction region between the particles exhibits a much higher LER than the opposing area, and the difference is especially evident from before 7 ns when the interparticle connection occurs.

From the findings so far, it can be concluded that the separation enhancement of PEDOT:PSS microphases under laser irradiation takes place in the following order. (1) Thermal energy induced by the laser selectively perturbs the $\pi-\pi$ interaction between PEDOT chains while preserving the electrostatic force between PEDOT and PSS^-. (2) Chain layers disintegrated from PEDOT-rich particles diffuse nondirectionally. However, PSSH matrices placed between adjacent particles are prone to structural rearrangements due to the high mechanical stress at the interface. This increases the possibility of particle-to-particle connectivity in a dynamic environment. (3) Once the bridges between the PEDOT-rich particles are formed, electrostatic interaction forces form in the direction of bringing the particles closer together. In other words, interparticle bridging formation is followed by interparticle integration and consequent further separation of PEDOT:PSS microphases. In addition, little physical mixing between PEDOT-rich and PSS-rich domains occurs throughout the entire process. Overall, the aforementioned process is remarkably similar to the coalescence observed in immiscible polymer blends.[49] The present study clarifies the manner in which a laser-induced thermodynamic driving force that initiates the phenomenon manifests molecular-scale phase changes.

4. CONCLUSIONS

This study explored the principle by which laser-induced energy supply imparts the electrical conductivity of PEDOT:PSS films in terms of nanoscale phase separation. The theoretical results using coarse-grained MD simulations demonstrate that the $\pi-\pi$ interactions and electrostatic forces of the PEDOT-rich phase are a key factor to inducing macroscopic coalescence of conductive cores. In particular, the laser-induced thermal energy selectively disrupted the layered structure of PEDOT molecules, ensuring high molecular mobility of the conductive particles while maintaining their inherent cohesion. This drives the physical separation between

https://doi.org/10.1021/acs.macromol.3c01870
Macromolecules 2024, 57, 2048−2056

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

the PEDOT-rich and PSS-rich phases in the laser irradiation process.

The initiation mechanism of the phase separation enhancement was further specified from the moment of connection between the PEDOT-rich particles. When energy is supplied by the laser, the mobility of the PSSH molecules around the PEDOT-rich particle increases collectively. Such deformation is more pronounced between two PEDOT-rich particles adjacent to each other, and the resulting mechanical stress exchange eventually works toward reducing the interparticle distance. Once connections are made between PEDOT-rich particles, the electrostatic interaction between PEDOT and PSS^- creates an attractive force that brings the particles closer together. Since the internal cohesion of PEDOT-rich particles remained high, such attraction leads to further separation of the PEDOT:PSS phase without destroying the particles or causing miscibility with the matrix.

Recently, the fabrication of high-performance PEDOT:PSS using laser-induced microannealing is of great engineering interest. The process is breaking new ground in fields requiring excellent aqueous stability and electrical properties, such as bioelectronics, skin-attached sensors, and energy storage devices. This work is the first effort to elucidate the practical effects of such laser-induced processes on the microstructure of materials from a theoretical perspective. It has been shown that the initiation process of PEDOT:PSS phase reorganization can be comprehended from evaluation of the mobility, micromorphology, and interaction forces of each molecular component in a high-energy supply environment. Therefore, the present findings can serve as a theoretical guidepost for the design of conductive polymers by laser annealing. Investigations on the laser-wavelength-specific behavior and the resulting coalescence dynamics, which can help further refine this technology, are currently underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.3c01870.

Experimental section; AFM images of PEDOT:PSS conductive films with increasing laser power; absorbance (left) and transmittance (right) spectra of the PEDOT:PSS and the pristine PSS films measured by the UV−vis spectroscopy; optical microscope images of PEDOT:PSS conductive films with increasing laser power; digital camera images of PEDOT:PSS conductive films with increasing laser power; energy supply in the simulation box equivalent to laser irradiation; key snapshots showing the process of forming an initial connection between two isolated PEDOT-rich particles; schematic for perimeter roughness measurement of PEDOT-rich particles (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Seung Hwan Ko − *Applied Nano and Thermal Science Lab, Department of Mechanical Engineering, Seoul National University, Seoul 08826, Republic of Korea;* orcid.org/0000-0002-7477-0820; Phone: +82-2-880-7114; Email: maxko@snu.ac.kr

Joonmyung Choi − *Department of Mechanical Design Engineering, Hanyang University, Seoul 04763, Republic of Korea; Department of Mechanical Engineering, BK21 FOUR ERICA-ACE Center, Hanyang University, Ansan 15588, Republic of Korea;* orcid.org/0000-0003-3673-8069; Phone: +82-31-400-5243; Email: joonchoi@hanyang.ac.kr

Authors

Hongdeok Kim − *Department of Mechanical Design Engineering, Hanyang University, Seoul 04763, Republic of Korea; Department of Mechanical Engineering, BK21 FOUR ERICA-ACE Center, Hanyang University, Ansan 15588, Republic of Korea*

Daeyeon Won − *Applied Nano and Thermal Science Lab, Department of Mechanical Engineering, Seoul National University, Seoul 08826, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.macromol.3c01870

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government(MSIT) (No. RS-2023-00210865 and No. 2021R1A2B5B03001691)

■ REFERENCES

(1) Tan, P.; Wang, H.; Xiao, F.; Lu, X.; Shang, W.; Deng, X.; Song, H.; Xu, Z.; Cao, J.; Gan, T.; et al. Solution-processable, soft, self-adhesive, and conductive polymer composites for soft electronics. *Nat. Commun.* **2022**, *13*, No. 358.

(2) Wang, Y.; Zhu, C.; Pfattner, R.; Yan, H.; Jin, L.; Chen, S.; Molina-Lopez, F.; Lissel, F.; Liu, J.; Rabiah, N. I.; et al. A highly stretchable, transparent, and conductive polymer. *Sci. Adv.* **2017**, *3*, No. e1602076.

(3) White, M. S.; Kaltenbrunner, M.; Głowacki, E. D.; Gutnichenko, K.; Kettlgruber, G.; Graz, I.; Aazou, S.; Ulbricht, C.; Egbe, D. A.; Miron, M. C.; et al. Ultrathin, highly flexible and stretchable PLEDs. *Nat. Photonics* **2013**, *7*, 811−816.

(4) Yuk, H.; Lu, B.; Lin, S.; Qu, K.; Xu, J.; Luo, J.; Zhao, X. 3D printing of conducting polymers. *Nat. Commun.* **2020**, *11*, No. 1604.

(5) Kim, N.; Lienemann, S.; Petsagkourakis, I.; Mengistie, D. A.; Kee, S.; Ederth, T.; Gueskine, V.; Leclère, P.; Lazzaroni, R.; Crispin, X.; Tybrandt, K. Elastic conducting polymer composites in thermoelectric modules. *Nat. Commun.* **2020**, *11*, No. 1424, DOI: 10.1038/s41467-020-15135-w.

(6) Yuk, H.; Lu, B.; Zhao, X. Hydrogel bioelectronics. *Chem. Soc. Rev.* **2019**, *48*, 1642−1667.

(7) Fan, X.; Nie, W.; Tsai, H.; Wang, N.; Huang, H.; Cheng, Y.; Wen, R.; Ma, L.; Yan, F.; Xia, Y. PEDOT:PSS for flexible and stretchable electronics: modifications, strategies, and applications. *Adv. Sci.* **2019**, *6*, No. 1900813.

(8) Shi, H.; Liu, C.; Jiang, Q.; Xu, J. Effective approaches to improve the electrical conductivity of PEDOT:PSS: a review. *Adv. Electron. Mater.* **2015**, *1*, No. 1500017.

(9) Fan, Z.; Ouyang, J. Thermoelectric properties of PEDOT:PSS. *Adv. Electron. Mater.* **2019**, *5*, No. 1800769.

(10) Won, D.; Kim, J.; Choi, J.; Kim, H.; Han, S.; Ha, I.; Bang, J.; Kim, K. K.; Lee, Y.; Kim, T.-S.; et al. Digital selective transformation and patterning of highly conductive hydrogel bioelectronics by laser-induced phase separation. *Sci. Adv.* **2022**, *8*, No. eabo3209.

(11) Lu, B.; Yuk, H.; Lin, S.; Jian, N.; Qu, K.; Xu, J.; Zhao, X. Pure pedot:Pss hydrogels. *Nat. Commun.* **2019**, *10*, No. 1043.

(12) Inoue, A.; Yuk, H.; Lu, B.; Zhao, X. Strong adhesion of wet conducting polymers on diverse substrates. *Sci. Adv.* **2020**, *6*, No. eaay5394.

Downloaded from pubs.acs.org/manobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026

(13) Roh, E.; Hwang, B.-U.; Kim, D.; Kim, B.-Y.; Lee, N.-E. Stretchable, transparent, ultrasensitive, and patchable strain sensor for human−machine interfaces comprising a nanohybrid of carbon nanotubes and conductive elastomers. *ACS Nano* **2015**, *9*, 6252−6261.
(14) Lang, U.; Müller, E.; Naujoks, N.; Dual, J. Microscopical investigations of PEDOT:PSS thin films. *Adv. Funct. Mater.* **2009**, *19*, 1215−1220.
(15) Lang, U.; Naujoks, N.; Dual, J. Mechanical characterization of PEDOT:PSS thin films. *Synth. Met.* **2009**, *159*, 473−479.
(16) Kemerink, M.; Timpanaro, S.; De Kok, M.; Meulenkamp, E.; Touwslager, F. Three-dimensional inhomogeneities in PEDOT:PSS films. *J. Phys. Chem. B* **2004**, *108*, 18820−18825.
(17) Timpanaro, S.; Kemerink, M.; Touwslager, F.; De Kok, M.; Schrader, S. Morphology and conductivity of PEDOT/PSS films studied by scanning−tunneling microscopy. *Chem. Phys. Lett.* **2004**, *394*, 339−343.
(18) Ouyang, J. Secondary doping" methods to significantly enhance the conductivity of PEDOT:PSS for its application as transparent electrode of optoelectronic devices. *Displays* **2013**, *34*, 423−436.
(19) Mengistie, D. A.; Wang, P.-C.; Chu, C.-W. Effect of molecular weight of additives on the conductivity of PEDOT:PSS and efficiency for ITO-free organic solar cells. *J. Mater. Chem. A* **2013**, *1*, 9907−9915, DOI: 10.1039/C3TA11726J.
(20) Yeon, C.; Kim, G.; Lim, J.; Yun, S. Highly conductive PEDOT:PSS treated by sodium dodecyl sulfate for stretchable fabric heaters. *RSC Adv.* **2017**, *7*, 5888−5897.
(21) Okuzaki, H.; Harashina, Y.; Yan, H. Highly conductive PEDOT/PSS microfibers fabricated by wet-spinning and dip-treatment in ethylene glycol. *Eur. Polym. J.* **2009**, *45*, 256−261.
(22) Lee, J. H.; Jeong, Y. R.; Lee, G.; Jin, S. W.; Lee, Y. H.; Hong, S. Y.; Park, H.; Kim, J. W.; Lee, S.-S.; Ha, J. S. Highly conductive, stretchable, and transparent PEDOT:PSS electrodes fabricated with triblock copolymer additives and acid treatment. *ACS Appl. Mater. Interfaces* **2018**, *10*, 28027−28035.
(23) Cruz-Cruz, I.; Reyes-Reyes, M.; Aguilar-Frutis, M. A.; Rodriguez, A.; López-Sandoval, R. Study of the effect of DMSO concentration on the thickness of the PSS insulating barrier in PEDOT:PSS thin films. *Synth. Met.* **2010**, *160*, 1501−1506.
(24) Lee, I.; Kim, G. W.; Yang, M.; Kim, T.-S. Simultaneously enhancing the cohesion and electrical conductivity of PEDOT:PSS conductive polymer films using DMSO additives. *ACS Appl. Mater. Interfaces* **2016**, *8*, 302−310.
(25) Yildirim, E.; Wu, G.; Yong, X.; Tan, T. L.; Zhu, Q.; Xu, J.; Ouyang, J.; Wang, J.-S.; Yang, S.-W. A theoretical mechanistic study on electrical conductivity enhancement of DMSO treated PEDOT:PSS. *J. Mater. Chem. C* **2018**, *6*, 5122−5131.
(26) Modarresi, M.; Zozoulenko, I. Why does solvent treatment increase the conductivity of PEDOT:PSS? Insight from molecular dynamics simulations. *Phys. Chem. Chem. Phys.* **2022**, *24*, 22073−22082.
(27) Yun, C.; Han, J. W.; Kim, S.; Lim, D. C.; Jung, H.; Lee, S.-H.; Jang, J.-W.; Yoo, S.; Leo, K.; Kim, Y. H. Generating semi-metallic conductivity in polymers by laser-driven nanostructural reorganization. *Mater. Horiz.* **2019**, *6*, 2143−2151.
(28) Pflug, T.; Anand, A.; Busse, S.; Olbrich, M.; Schubert, U. S.; Hoppe, H.; Horn, A. Spatial conductivity distribution in thin PEDOT:PSS films after laser microannealing. *ACS Appl. Electron. Mater.* **2021**, *3*, 2825−2831.
(29) Ding, C.-F.; Lan, Y.-T.; Young, H.-T.; Hsiao, W.-T. Enhanced conductivity of the Poly (3, 4-ethylenedioxythiophene)-poly (styrenesulfonate)(PEDOT:PSS) films using pulsed CO2 laser irradiation. *Synth. Met.* **2021**, *275*, No. 116743.
(30) Troughton, J.; Peillon, N.; Borbely, A.; Rodriguez-Pereira, J.; Pavlinak, D.; Macak, J. M.; Djenizian, T.; Ramuz, M. High conductivity PEDOT:PSS through laser micro-annealing: Mechanisms and application. *J. Mater. Chem. C* **2022**, *10*, 16592−16603.
(31) Wang, X.; Feng, Y.; Sun, K.; Chai, N.; Mai, B.; Li, S.; Chen, X.; Zhao, W.; Zhang, Q. Ultrafast Laser-Induced Excellent Thermoelectric Performance of PEDOT:PSS Films. *Energy Environ. Mater.* **2023**, No. e12650.
(32) Volkov, A. V.; Wijeratne, K.; Mitraka, E.; Ail, U.; Zhao, D.; Tybrandt, K.; Andreasen, J. W.; Berggren, M.; Crispin, X.; Zozoulenko, I. V. Understanding the capacitance of PEDOT:PSS. *Adv. Funct. Mater.* **2017**, *27*, No. 1700329.
(33) Crispin, X.; Jakobsson, F.; Crispin, A.; Grim, P.; Andersson, P.; Volodin, Av.; Van Haesendonck, C.; Van der Auweraer, M.; Salaneck, W. R.; Berggren, M. The origin of the high conductivity of poly (3, 4-ethylenedioxythiophene)− poly (styrenesulfonate)(PEDOT− PSS) plastic electrodes. *Chem. Mater.* **2006**, *18*, 4354−4360.
(34) Ouyang, J.; Xu, Q.; Chu, C.-W.; Yang, Y.; Li, G.; Shinar, J. On the mechanism of conductivity enhancement in poly (3, 4-ethylenedioxythiophene): poly (styrene sulfonate) film through solvent treatment. *Polymer* **2004**, *45*, 8443−8450.
(35) Chiu, W. W.; Travaš-Sejdić, J.; Cooney, R. P.; Bowmaker, G. A. Studies of dopant effects in poly (3, 4-ethylenedi-oxythiophene) using Raman spectroscopy. *J. Raman Spectrosc.* **2006**, *37*, 1354−1361.
(36) Moreno, J. M. C.; Yoshimura, M. Hydrothermal processing of high-quality multiwall nanotubes from amorphous carbon. *J. Am. Chem. Soc.* **2001**, *123*, 741−742, DOI: 10.1021/ja003008h.
(37) Kim, S.-M.; Kim, C.-H.; Kim, Y.; Kim, N.; Lee, W.-J.; Lee, E.-H.; Kim, D.; Park, S.; Lee, K.; Rivnay, J.; Yoon, M. H. Influence of PEDOT:PSS crystallinity and composition on electrochemical transistor performance and long-term stability. *Nat. Commun.* **2018**, *9*, No. 3858.
(38) Hu, A.; Tan, L.; Hu, X.; Hu, L.; Ai, Q.; Meng, X.; Chen, L.; Chen, Y. Crystallization and conformation engineering of solution-processed polymer transparent electrodes with high conductivity. *J. Mater. Chem. C* **2017**, *5*, 382−389.
(39) Marrink, S. J.; Risselada, H. J.; Yefimov, S.; Tieleman, D. P.; De Vries, A. H. The MARTINI force field: coarse grained model for biomolecular simulations. *J. Phys. Chem. B* **2007**, *111*, 7812−7824.
(40) Modarresi, M.; Mehandzhiyski, A.; Fahlman, M.; Tybrandt, K.; Zozoulenko, I. Microscopic Understanding of the Granular Structure and the Swelling of PEDOT:PSS. *Macromolecules* **2020**, *53*, 6267−6278.
(41) Modarresi, M.; Franco-Gonzalez, J. F.; Zozoulenko, I. Morphology and ion diffusion in PEDOT: Tos. A coarse grained molecular dynamics simulation. *Phys. Chem. Chem. Phys.* **2018**, *20*, 17188−17198.
(42) Vögele, M.; Holm, C.; Smiatek, J. Coarse-grained simulations of polyelectrolyte complexes: MARTINI models for poly (styrene sulfonate) and poly (diallyldimethylammonium). *J. Chem. Phys.* **2015**, *143*, No. 243151.
(43) Franco-Gonzalez, J. F.; Zozoulenko, I. V. Molecular dynamics study of morphology of doped PEDOT: from solution to dry phase. *J. Phys. Chem. B* **2017**, *121*, 4299−4307.
(44) Xiao, L.; Chen, X.; Yang, X.; Sun, J.; Geng, J. Recent Advances in Polymer-Based Photothermal Materials for Biological Applications. *ACS Appl. Polym. Mater.* **2020**, *2*, 4273−4288.
(45) Mahajan, M. S.; Marathe, D.; Ghosh, S. S.; Ganesan, V.; Sali, J. V. Changes in in-plane electrical conductivity of PEDOT:PSS thin films due to electric field induced dipolar reorientation. *RSC Adv.* **2015**, *5*, 86393−86401.
(46) Lim, K.; Kang, J.; Jung, S.; Lee, S.; Park, J.; Kim, D. G.; Kang, Y. C. Improving electrical conductivity of PEDOT:PSS with phase separation by applying electric fields. *Bull. Korean Chem. Soc.* **2018**, *39*, 469−476.
(47) Alessandri, R.; Uusitalo, J. J.; De Vries, A. H.; Havenith, R. W.; Marrink, S. J. Bulk heterojunction morphologies with atomistic resolution from coarse-grain solvent evaporation simulations. *J. Am. Chem. Soc.* **2017**, *139*, 3697−3705.
(48) Heyes, D. M. Pressure tensor of partial-charge and point-dipole lattices with bulk and surface geometries. *Phys. Rev. B* **1994**, *49*, No. 755, DOI: 10.1103/PhysRevB.49.755.
(49) Fortelný, I.; Jůza, J.; Dimzoski, B. Coalescence in quiescent polymer blends with a high content of the dispersed phase. *Eur. Polym. J.* **2012**, *48*, 1230−1240.

Downloaded from pubs.acs.org/mamobx/article-pdf/57/5/2048/1389648/ma3c01870.pdf by SUNGKYUNKWAN UNIV user on 11 October 2026